

Paper review:

Causal inference on distribution functions

Zhenhua Lin, Dehan Kong and Linbo Wang (2023)

Suehyun Kim

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Causal Inference Lab.
Seoul National University

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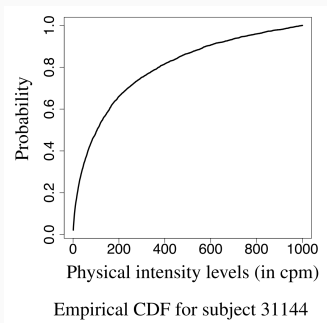
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Introduction

The aim of this paper is to develop a novel framework for causal inference with **distribution-valued outcomes**.



How can the notion of a causal effect be defined for distributions?

The aim of this paper is to develop a novel framework for causal inference with **distribution-valued outcomes**.

We will look at:

- The definitions of mean potential outcomes and average causal effect, based on the properties of the 2-Wasserstein space.
- *How* and *why* they are defined in this way.
- The identification and estimation of the causal effect, and the asymptotic properties of the proposed estimators.
- Simulation studies and a data application.

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OT and Wasserstein space: A brief introduction

The **Optimal Transport (OT)** problem studies how to move mass from one probability distribution to another at minimal cost.

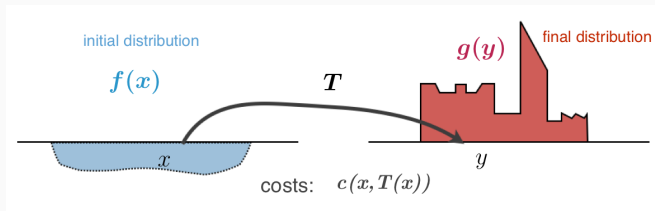
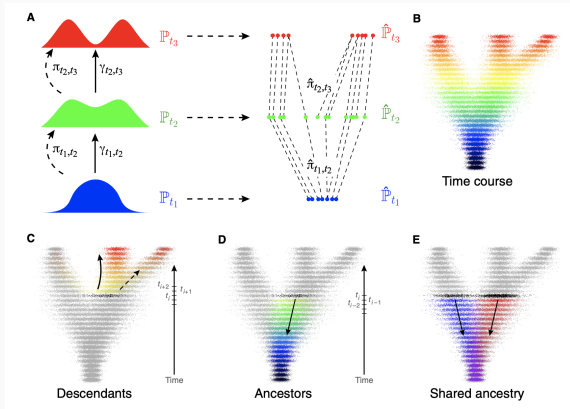


Figure reference: <https://www.fv-berlin.de/en/research/research-highlight/von-a-nach-b-oder-doch-besser-nach-c-en>

The **Wasserstein space** arises naturally as the metric space induced by the optimal transport problem – OT provides the geometry underlying distances between probability measures.

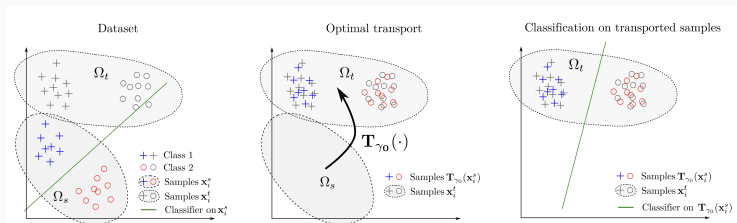
OT and Wasserstein space: A brief introduction

Example 1. Cellular differentiation (Schiebinger et al., 2019)



OT and Wasserstein space: A brief introduction

Example 2. Optimal transport for domain adaptation (Courty et al., 2017)



OT and Wasserstein space: A brief introduction

The Monge problem (1781). *Given a pile of sand and a pit of equal volume, how can one optimally transport the sand into the pit?*

- Suppose there is a ‘sand space’ $\mathcal{X}$, a ‘pit space’ $\mathcal{Y}$, and a cost function $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ that encapsulates the cost of transporting a unit mass from $x \in \mathcal{X}$ to $y \in \mathcal{Y}$.
- Let μ and ν be two probability measures on $\mathcal{X}$ and $\mathcal{Y}$, respectively, representing the distribution of sand and the shape of the pit.
- Our decision where to put each unit of sand can be thought of as a function $T : \mathcal{X} \rightarrow \mathcal{Y}$, which incurs a total transport cost of

$$C(T) = \int_{\mathcal{X}} c(x, T(x)) \, d\mu(x).$$

OT and Wasserstein space: A brief introduction

The Monge problem (1781).

- The map T must be *mass-preserving*:

$$\mu(T^{-1}(B)) = \nu(B), \quad \forall B \in \mathcal{B}(\mathcal{Y})$$

- In other words, ν is the pushforward measure of μ by T , denoted as $T\#\mu = \nu$.
- The **Monge problem** is to find the optimal transport map, that is, to solve

$$\inf_{T: T\#\mu = \nu} C(T).$$

- However, the Monge problem is very difficult, as the set of transport maps $\{T : T\#\mu = \nu\}$ is intractable, and the constraint $T\#\mu = \nu$ is highly non-linear.

The Monge-Kantorovich problem (1942). In 1942, Kantorovich proposed a relaxation of the Monge problem, in which mass can be split.

- Instead of assigning a fixed destination $T(x)$ to each $x \in \mathcal{X}$, one constructs a probability measure μ_x that describes how the mass at x is distributed in $\mathcal{Y}$.
- Formally, a *transference plan* (or a *coupling*) is a probability measure π on $\mathcal{X} \times \mathcal{Y}$ whose marginals are μ and ν :

$$\pi(A \times \mathcal{Y}) = \mu(A), \pi(\mathcal{X} \times B) = \nu(B), \forall A \in \mathcal{B}(\mathcal{X}), B \in \mathcal{B}(\mathcal{Y}).$$

Denote the set of all couplings of μ and ν by $\Pi(\mu, \nu)$.

The Monge-Kantorovich problem (1942).

- Then, the total cost associated with $\pi \in \Pi(\mu, \nu)$ is

$$C(\pi) = \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \, d\pi(x, y).$$

- The **Monge-Kantorovich problem** is to find the optimal transference plan, that is, to solve

$$\inf_{\pi \in \Pi(\mu, \nu)} C(\pi).$$

- The Monge-Kantorovich problem is always feasible, and a minimiser is guaranteed to exist under mild conditions.

The one-dimensional case

When $\mathcal{X} = \mathcal{Y} = \mathbb{R}$, the Monge-Kantorovich problem has a particularly simple structure.

- Identify $\mu, \nu \in P(\mathbb{R})$ with their cdfs F and G .
- If the transport map T is increasing and $T\#\mu = \nu$, then for all $x \in \mathbb{R}$,

$$G(t) = \nu((-\infty, t]) = \mu((-\infty, T^{-1}(t)]) = F(T^{-1}(t)),$$

which implies that $T = G^{-1} \circ F$ is uniquely determined.

- If F is discontinuous, then T is not necessarily defined. However, there exists an optimal coupling in an analogous sense.

The one-dimensional case

Theorem (Optimal transport in $\mathbb{R}$)

Let $\mu, \nu \in P(\mathbb{R})$ with distribution functions F and G , respectively, and let the cost function be of the form $c(x, y) = h(|x - y|)$, where h is nonnegative and convex. Then

$$\inf_{\pi \in \Pi(\mu, \nu)} C(\pi) = \int_0^1 h(G^{-1}(u) - F^{-1}(u)) \, du.$$

- If the infimum is finite and h is strictly convex, then the optimal transference plan is unique.
- Furthermore, if F is continuous, then the infimum is attained by the transport map $T = G^{-1} \circ F$.

The Monge-Kantorovich problem gives rise to a metric structure, the **Wasserstein distance**, in the space of probability measures $P(\mathcal{X})$ on a space $\mathcal{X}$.

Definition (p -Wasserstein space)

Let $\mathcal{X}$ be a separable Banach space. For $p \geq 1$, the *p -Wasserstein space* on $\mathcal{X}$ is defined as

$$\mathcal{W}_p(\mathcal{X}) = \left\{ \mu \in P(\mathcal{X}) : \int_{\mathcal{X}} \|x\|^p d\mu(x) < \infty \right\}.$$

Wasserstein space

The p -Wasserstein space is a metric space equipped with the p -**Wasserstein distance** $W_p(\mu, \nu)$:

$$W_p(\mu, \nu) := \left(\inf_{\pi \in \Pi(\mu, \nu)} C_p(\pi) \right)^{1/p} = \left\{ \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X}^2} \|x_1 - x_2\|^p d\pi(x_1, x_2) \right\}^{1/p}.$$

- In words, $W_p(\mu, \nu)$ is defined as the *minimal total transportation cost* between μ and ν wrt the cost function $c(x_1, x_2) = \|x_1 - x_2\|^p$.
- We focus on the 2-Wasserstein space $\mathcal{W}_2(\mathcal{I})$, where $p = 2$ and $\mathcal{X} = \mathcal{I} \subseteq \mathbb{R}$ is a bounded interval.
- In this case, the one-dimensional optimal transport theorem implies that

$$W_2(\mu, \nu) = \|F_\mu^{-1} - F_\nu^{-1}\|_{L^2[0,1]},$$

so that $\mu \mapsto F_\mu^{-1}$ is an *isometry* from $\mathcal{W}_2(\mathcal{I})$ into a subset of $L^2[0, 1]$.

The concept of a **Fréchet mean** generalizes the notion of an average to metric spaces:

- If we define $F(\theta) = \sum_{i=1}^N \|\theta - x_i\|^2$, then $\theta = \bar{x}_N$ is the minimizer of $F(\theta)$.
- The Fréchet mean is defined by replacing the usual *sum of squares* with the *sum of squared distances*.
- In analysis, the Fréchet mean is also known as the **barycentre**.

Definition (Empirical Fréchet functional and mean)

The Fréchet functional associated with measures $\nu_1, \dots, \nu_N \in \mathcal{W}_2(\mathcal{X})$ is defined as

$$F(\nu) = \frac{1}{N} \sum_{i=1}^N W_2^2(\nu, \nu_i), \quad \nu \in \mathcal{W}_2(\mathcal{X}).$$

A *Fréchet mean* (or *Wasserstein barycentre*) is a minimizer of F in $\mathcal{W}_2(\mathcal{X})$ (if it exists).

Remark. Due to the isometry between the 2-Wasserstein space and the space of quantile functions in $\mathbb{R}$, the Wasserstein barycenter of $\mu^1, \dots, \mu^N \in \mathcal{W}_2(\mathbb{R})$ is the measure μ with the quantile function

$$F_\mu^{-1} = \frac{1}{N} \sum_{i=1}^N F_{\mu^i}^{-1}.$$

The Wasserstein barycentre and distance have several features that make them appealing for defining causal effects on distribution-valued data:

- (i) The Wasserstein barycentre *reduces to the usual Euclidean mean* in the degenerate case where the distributions are point masses.
- (ii) Compared with the Euclidean distance and other distances such as the Hellinger distance, the Wasserstein barycentre performs exceptionally well in *capturing the structure of random distributions*.
- (iii) The Wasserstein distance has an intuitive interpretation of the amount of work needed to transform one distribution to another, and it comes with a map that shows how to move from one random object to another.

Wasserstein barycentre

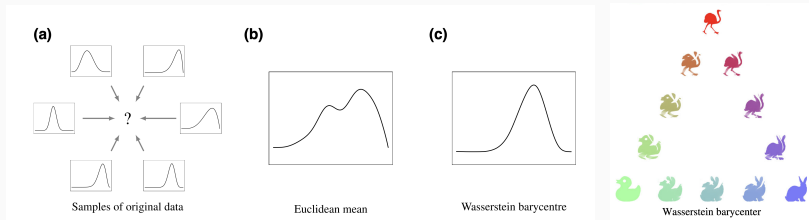


Figure 1 (left). Comparison of Wasserstein barycentre vs. Euclidean average for capturing structural information in random distributions. The Euclidean average has two modes while the Wasserstein barycentre preserves unimodality.

Figure (right). Bonneel et al. (2015)

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We use the well-known potential outcomes framework.

- $A \in \{0, 1\}$: Binary treatment indicator
- $X \in \mathbb{R}^d$: Baseline covariates
- $Y(a)$: Potential outcomes; *a random object that resides in a possibly nonlinear space*
- Y : Observed outcome
- We observe n independent samples (A_i, X_i, Y_i) , $i = 1, \dots, n$ from an infinite super-population of (A, X, Y) .

Assumptions: SUTVA, strong ignorability, positivity.

Definition of causal effects

In the familiar case where $Y(a)$ resides in $\mathbb{R}$, the causal effect is commonly defined as the contrast between a summary measure of the potential outcome distributions.

- **ACE (Average causal effect)**

$$\text{ACE} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

- **QTE (Quantile treatment effect)**

$$\text{QTE}(\alpha) = F_{Y(1)}^{-1}(\alpha) - F_{Y(0)}^{-1}(\alpha),$$

where $F_Z(z) = P(Z \leq z)$ is the cdf of the random variable Z , and $F_Z^{-1}(\alpha)$ is the corresponding quantile function.

Definition of causal effects

Causal effects defined in this manner can be interpreted on the *population level*, as they concern contrasts between potential outcomes in two hypothetical populations.

There is, however, a subtle but important distinction in the *individual-level interpretation*:

- Since $ACE = \mathbb{E}[CE_i]$, the ACE can also be interpreted as averages of individual causal effects.
- However, $QTE(\alpha)$ cannot be interpreted as the α -quantile of individual causal effects.

Definition of causal effects in $\mathcal{W}_2(\mathcal{I})$

We wish to generalize such notions in $\mathcal{W}_2(\mathcal{I})$. First, we define the **mean potential outcomes** using their Wasserstein barycentres:

Definition (mean potential outcome)

For $Y(a) \in \mathcal{W}_2(\mathcal{I})$, we define the mean potential outcomes

$$\mu_a = \mathbb{E}Y(a) := \operatorname{argmin}_{\nu \in \mathcal{W}_2(\mathcal{I})} \mathbb{E}[W_2^2(Y(a), \nu)], \quad a = 0, 1.$$

i.e. μ_a is a 'typical' potential distribution under treatment $A = a$.

But what about the *causal effect*? Do we simply take the difference as we did in $\mathbb{R}$?

Definition of causal effects in $\mathcal{W}_2(\mathcal{I})$

Ideally, a causal effect definition in the Wasserstein space should satisfy the following desiderata:

- (a) When $\mathbb{E}Y(1) = \mathbb{E}Y(0)$, the causal effect equals zero.
- (b) In the degenerate case where $Y_i(a) = \delta_{y_i(a)}$, corresponding to the classical scenario in $\mathbb{R}$, the causal effect is identical to the usual ACE.
- (c) The average causal effect is a *contrast between the averages of potential outcomes in two hypothetical populations*, and thus can be interpreted on the population level.
- (d) The average causal effect equals the *average of individual causal effects*, thus maintaining the individual-level interpretation of the ACE for real-valued outcomes.

Definition of causal effects in $\mathcal{W}_2(\mathcal{I})$

The following novel definition of average causal effect, namely the **causal effect map**, satisfies all desiderata (a) - (d).

Definition 1 (causal effect map)

Let λ be a continuous distribution function (*reference distribution*).

The **individual causal effect map** of A on Y is defined as

$$\Delta_i^\lambda(\cdot) = Y_i(1)^{-1} \circ \lambda(\cdot) - Y_i(0)^{-1} \circ \lambda(\cdot).$$

The **(average) causal effect map** is defined as

$$\Delta^{\lambda(\cdot)} = (\mathbb{E}Y_i(1))^{-1} \circ \lambda(\cdot) - (\mathbb{E}Y_i(0))^{-1} \circ \lambda(\cdot) = (\mu_1^{-1} - \mu_0^{-1}) \circ \lambda(\cdot).$$

Definition of causal effects in $\mathcal{W}_2(\mathcal{I})$

The *reference distribution* λ plays a key role in the definition of the causal effect map.

- It is allowed to have a domain different from that of Y .
- Different choices of λ lead to different interpretations of the causal effect maps.

Interpretation 1 (Difference in quantiles)

If the reference distribution λ is the *uniform distribution* on $[0, 1]$ so that $\lambda(t) = t$, $t \in [0, 1]$, then the causal effect map $\Delta(\cdot)$ can be interpreted as *difference in quantiles*.

Definition of causal effects in $\mathcal{W}_2(\mathcal{I})$

Analogous to the concepts of optimal transport maps, we can define *causal transport maps*:

- **Individual causal transport map:** $T_i(\cdot) = Y_i(1)^{-1} \circ Y_i(0)(\cdot)$
- **Causal transport map:** $T(\cdot) = \mu_1^{-1} \circ \mu_0(\cdot)$.

Interpretation 2 (Causal transport maps)

Suppose $Y(a)$ are continuous. If the reference distribution is chosen to be the barycentres μ_0 or μ_1 , then

$$\Delta^{\mu_0} = T - \text{id} = \mu_1^{-1} \circ \mu_0 - \text{id}, \quad \Delta^{\mu_1} = \text{id} - T^{-1} = \text{id} - \mu_0^{-1} \circ \mu_1.$$

If the reference distribution is chosen to be Y_i , then

$$\Delta_i^{Y_i} = \begin{cases} T_i - \text{id} = Y_i(1)^{-1} \circ Y_i(0) - \text{id} & \text{if } A_i = 0 \\ \text{id} - T_i^{-1} = \text{id} - Y_i(0)^{-1} \circ Y_i(1) & \text{if } A_i = 1. \end{cases}$$

Properties of causal effect maps

We now describe several desirable properties of the causal effect maps.

- Desiderata (a) - (c) can be verified easily.
- The following theorem demonstrates desideratum (d), which also plays a crucial role in the identification procedure.

Theorem 1

The *average causal effect map corresponds to the average of individual causal effect maps* with respect to the same reference distribution λ :

$$\Delta^\lambda(\cdot) = \mathbb{E}\Delta_i^\lambda(\cdot) = \mathbb{E}[Y(1)^{-1} \circ \lambda(\cdot) - Y(0)^{-1} \circ \lambda(\cdot)].$$

Remark. In contrast, the (population) causal transport map is generally different from the average of individual causal transport maps, i.e.

$$T \neq \mathbb{E}[T_i].$$

Properties of causal effect maps

In some scenarios, practitioners may want to compute the *Wasserstein distance* $W_2(\mu_1, \mu_0)$, which is a scalar quantity that can be interpreted as the magnitude of the causal effect.

The following proposition shows how to compute $W_2(\mu_1, \mu_0)$ from $\Delta^\lambda(\cdot)$.

Proposition 2

The Wasserstein distance $W_2(\mu_1, \mu_0)$ is determined by the causal effect map $\Delta^\lambda(\cdot)$ via

$$W_2(\mu_1, \mu_0) = \|\Delta^\lambda\|_\lambda = \left\{ \int (\Delta^\lambda)^2(u) d\lambda(u) \right\}^{1/2}.$$

Identification. Under the ignorability and positivity assumptions, we can identify the causal effect map Δ^λ .

Theorem 2

The average causal effect map Δ^λ is identifiable and given by

$$\Delta^\lambda = \mu_1^{-1,\lambda} - \mu_0^{-1,\lambda}$$

where

$$\mu_a^{-1,\lambda} = \mathbb{E}_X[\mathbb{E}[(Y^{-1} \circ \lambda) \mid A = a, X]] = \mathbb{E}\left[\frac{I(A = a)(Y^{-1} \circ \lambda)}{P(A = a \mid X)}\right].$$

Remarks.

- In general, when $Y(a)$ is a random distribution function,

$$\mu_a \neq \mathbb{E}[\mathbb{E}[Y \mid A = a, X]] = \mathbb{E} \left[\frac{I(A = a)Y}{P(A = a \mid X)} \right],$$

so the mean potential outcomes μ_a are not directly identifiable.

- Instead, μ_a *may be identified as the inverse of $\mu_a^{-1, \lambda} \circ \lambda^{-1}$* , using Theorem 2.

Estimation. We construct two estimators for the causal effect map Δ^λ :
(i) *the doubly robust estimator*, and (ii) *the cross-fitting estimator*.

Notations:

- $m_a^\lambda(X) = \mathbb{E}[Y^{-1} \circ \lambda \mid A = a, X]$ (conditional outcome)
- $\pi(X) = P(A = 1 \mid X)$ (propensity score)
- $f(A \mid X) = A\pi(X) + (1 - A)(1 - \pi(X))$

Identification and estimation

Note that, in practice, the outcome Y_i itself might not be fully observed. Hence, we construct estimators according to the following steps:

Step 1. Use standard nonparametric methods to obtain the estimates $\hat{Y}_i$.

- Nonparametric MLE, local polynomial smoothing

Step 2. If the reference distribution λ is unknown, obtain its estimate $\hat{\lambda}$.

Step 3. Regress $\hat{Y}^{-1} \circ \hat{\lambda}$ on X and A to obtain an estimate of $m_a^\lambda(X)$ for each individual, denoted as $\hat{m}_a^\lambda(X_i)$.

- Standard functional regression models

Step 4. Construct an estimate $\hat{f}(A | X)$ of $f(A | X)$.

- PS models (logistic regression, etc.)

Step 5. Construct the DR and CF estimators.

Estimator 1 (Doubly robust estimator)

The doubly robust estimator for Δ^λ is constructed as

$$\hat{\Delta}_{DR}^\lambda = \hat{\mu}_1^{-1,\lambda} - \hat{\mu}_0^{-1,\lambda}$$

where

$$\hat{\mu}_a^{-1,\lambda} = \mathbb{P}_n \left[\hat{m}_a^{\hat{\lambda}}(X) + \frac{I(A=a)}{\hat{f}(A|X)} \left\{ \hat{Y}^{-1} \circ \hat{\lambda} - \hat{m}_a^{\hat{\lambda}}(X) \right\} \right].$$

i.e. it combines an outcome model with an IPW estimator, like the DR estimator in $\mathbb{R}$.

Estimator 2 (Cross-fitting estimator)

The cross-fitting estimator is constructed as follows:

- (i) Partition the data into K parts of roughly equal sizes, denoted by $\mathcal{D}_1, \dots, \mathcal{D}_K$.
- (ii) For $k = 1, \dots, K$, use $\mathcal{D}_{-k} = \cup_{m \neq k} \mathcal{D}_m$ to obtain estimates $\hat{f}_k, \hat{\pi}_k, \hat{m}_{a,k}^{\hat{\lambda}_k}$ and $\hat{\lambda}_k$.
- (iii) In each partition, use $\mathcal{D}_k$ to estimate the causal effect by $\Delta_{CF,k}^{\hat{\lambda}_k} = \hat{\mu}_{1,k}^{-1, \hat{\lambda}_k} - \hat{\mu}_{0,k}^{-1, \hat{\lambda}_k}$, where

$$\hat{\mu}_{a,k}^{-1, \hat{\lambda}_k} = n_k^{-1} \sum_{i \in \mathcal{D}_k} \left[\hat{m}_{a,k}^{\hat{\lambda}_k}(X_i) + \frac{I(A_i = a)}{\hat{f}_k(A_i | X_i)} \left\{ \hat{Y}_i^{-1} \circ \hat{\lambda}_k - \hat{m}_{a,k}^{\hat{\lambda}_k}(X_i) \right\} \right].$$

Estimator 2 (Cross-fitting estimator (continued))

(iv) Combine the effects from different partitions via

$$\hat{\mu}_{a,CF}^{-1,\hat{\lambda}} = \sum_{k=1}^K \frac{n_k}{n} \hat{\mu}_{a,k}^{-1,\hat{\lambda}_k} \circ \hat{\lambda}_k^{-1} \circ \hat{\lambda} \quad \text{and} \quad \hat{\Delta}_{CF}^{\hat{\lambda}} = \hat{\mu}_{1,CF}^{-1,\hat{\lambda}} - \hat{\mu}_{0,CF}^{-1,\hat{\lambda}},$$

where $\hat{\lambda}$ is estimated using the whole dataset.

If the reference distribution $\hat{\lambda}_k$ is fixed to a common distribution $\hat{\lambda}$, the above is reduced to $\hat{\Delta}_{CF}^{\hat{\lambda}} = n^{-1} \sum_{k=1}^K n_k \Delta_{CF,k}^{\hat{\lambda}}$.

Remark. In the DR procedure, the data are used twice – once for estimating the nuisance functions, and the other for the causal effect – which hinders theoretical analyses of the estimator. The CF estimator avoids such an issue (Chernozhukov et al., 2018).

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To derive the asymptotic properties of the proposed estimators, we make the following assumptions:

- (A1) The estimates $\hat{Y}_i$ are independent, and converge to their population counterparts at appropriate rates.
- (A2) $W_2(\hat{\lambda}, \lambda)$ converges to zero at an appropriate rate.
- (A3) The OR and PS estimates converge, and the estimated PS is bounded away from 0 and 1.
- (A4) The OR estimate $\hat{m}_a^{\hat{\lambda}}$ is close to the estimate $\tilde{m}_a^{\lambda}$, which is obtained using the true reference distribution λ .

For the DR estimator, we need the following additional assumption:

(A5) For a real number $t \in \mathcal{T}$, consider real-valued functions of the form

$$(t \times \check{m}^\lambda \times \check{\pi})(A, X, Y) = \frac{(aA + (1-a)(1-A))\{(Y^{-1} \circ \lambda)(t) - \check{m}_a^\lambda(X)(t)\}}{a\check{\pi}(X) + (1-a)(1 - \check{\pi}(X))} + \check{m}_a^\lambda(X)(t)$$

defined on (A, X, Y) .

Let $\mathcal{F}_a^\lambda$ be a class of such functions, and let $\mathcal{M}_a^\lambda$ be a class of functions containing m_a^λ . Then,

- The OR and PS model estimates are stable, in the sense that each data point has asymptotically equal contribution to the estimators.
- $\mathcal{F}_a^\lambda$ is a Donsker class, and the complexity of $\mathcal{M}_a^\lambda$ is bounded.

Theorem 3 (Asymptotic properties of the DR estimator)

Suppose that both $\hat{\lambda}$ and λ are continuous distribution functions, and assumptions (A1) - (A5) hold. Then,

- (i) $\|\hat{\Delta}_{DR}^{\hat{\lambda}} \circ \hat{\lambda}^{-1} \circ \lambda - \Delta^{\lambda}\|_{\lambda} = O_P(n^{-1/2} + n^{-1/2} \varrho_m^{1/2} + n^{-1/2} \varrho_{\pi} + \varrho_m \varrho_{\pi});$
- (ii) if $\varrho_m \varrho_{\pi} = o(n^{-1/2})$, $\varrho_m = o(1)$, and $\varrho_{\pi} = o(1)$, then

$$\sqrt{n}(\hat{\Delta}_{DR}^{\hat{\lambda}} \circ \hat{\lambda}^{-1} \circ \lambda - \Delta^{\lambda}) = \sqrt{n}(\mathbb{P}_n - \mathbb{E})\{\phi(A, X, Y)\} + o_P(1),$$

thus it converges weakly to a centered Gaussian process in $L^2(\mathcal{J}; \lambda)$ with the same asymptotic distribution as $\sqrt{n}(\mathbb{P}_n - \mathbb{E})\{\phi(A, X, Y)\}$.

Notations:

- $\varrho_{\pi}^4 := \mathbb{E}\|\hat{\pi} - \pi\|_2^4$, $\varrho_m^4 := \max\{\mathbb{E}\|\tilde{m}_0^{\lambda} - m_0^{\lambda}\|_{\lambda}^4, \mathbb{E}\|\tilde{m}_1^{\lambda} - m_1^{\lambda}\|_{\lambda}^4\}$
- $\phi(A, X, Y) = \frac{A\{Y^{-1} \circ \lambda - m_1^{\lambda}(X)\}}{\pi(X)} + m_1^{\lambda}(X) - \frac{(1-A)\{Y^{-1} \circ \lambda - m_0^{\lambda}(X)\}}{1 - \pi(X)} - m_0^{\lambda}(X)$

Remarks.

- **Theorem 3** shows that $\hat{\Delta}_{DR}^{\hat{\lambda}}$ enjoys the double robustness property, and that $\hat{\Delta}_{DR}^{\hat{\lambda}}$ is an asymptotically linear estimator with influence function $\phi(A, X, Y) - \mathbb{E}\phi(A, X, Y)$.
- The covariance function of the limiting Gaussian process can be estimated from data, using the estimates of the OR and PS models.
- Based on the estimated covariance function and the asymptotic normality result, we can construct *approximate simultaneous confidence bands (SCBs)* for Δ^{λ} via a resampling strategy.
- Assumption (A5) is rather technical, and may fail in settings invoking machine learning methods. The CF estimator does not require this assumption, and has analogous asymptotic properties (see **Theorem 4** in the original paper).

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Simulation 1

- Performance of the estimators (bias & SE)
- Effectiveness of the doubly robust property
- Coverage probability of the simultaneous confidence bands

Simulation 2

- Performance of data-adaptive DR and CF estimators when the outcome and PS models are nonlinear

Simulation setup

Data generation: The simulation data consist of n independent samples from the joint distribution of (A, X, Y) .

- $n = 50, 200, 1,000$
- Distributions of (A, X, Y) and ϵ :
 - $X \sim \text{Unif}[0, 1]$
 - $P(A = 1 \mid X) = \text{expit}(1 + X)$
 - $Y^{-1}(\alpha) = (-\mathbb{E}(A) + A + X + \epsilon) \sin(\pi\alpha)/8 + \alpha$
(note that $\mathcal{I} = [0, 1]$ and $Y \in \mathcal{W}_2(\mathcal{I})$)
 - $\epsilon \sim \text{Unif}[-0.5, 0.5]$
- For each subject $i = 1, \dots, n$, assume that 1,001 i.i.d. observations are available from the distribution function $Y_i(t)$.
 - The outcomes $Y_i(t)$ are estimated using the empirical cdf + linear interpolation.

Target estimand: *Causal effect map at the reference distribution μ_0 .*

- True value: $\Delta^{\mu_0}(t) = \sin(\pi\mu_0(t))/8$

Model specification:

- Outcome: Correct linear model (with predictor X) / Incorrect (X^2)
- PS: Correct logistic model (with predictor X) / Incorrect (X^2)

Evaluation metrics:

- Bias of difference in medians: $\hat{\Delta}^{\hat{\mu}_0}(\hat{\mu}_0^{-1}(0.5)) - \Delta^{\mu_0}(\mu_0^{-1}(0.5))$
- Root mean integrated squared error under the reference distribution μ_0 :
$$\|\hat{\Delta}^{\hat{\mu}_0} \circ \hat{\mu}_0^{-1} \circ \mu_0 - \Delta^{\mu_0}\|_{\mu_0}$$

Simulation 1: General performance

Results. 1,000 Monte Carlo replicates for each setting, compared against OR and IPW estimators.

Estimator	Model		Sample size		
	PS	OR	$n = 50$	$n = 200$	$n = 1,000$
Bias of difference in medians $\times 100$					
$\hat{\Delta}_{OR}^{\beta_0}$	–	✓	0.010(0.037)	–0.036(0.018)	0.008(0.008)
$\hat{\Delta}_{OR}^{\beta_0}$	–	×	3.898(0.075)	3.787(0.037)	3.819(0.017)
$\hat{\Delta}_{IPW}^{\beta_0}$	✓	–	0.433(0.157)	0.008(0.045)	0.021(0.018)
$\hat{\Delta}_{IPW}^{\beta_0}$	×	–	3.975(0.084)	3.706(0.036)	3.738(0.016)
$\hat{\Delta}_{DR}^{\beta_0}$	✓	✓	0.005(0.04)	–0.033(0.019)	0.013(0.008)
$\hat{\Delta}_{DR}^{\beta_0}$	✓	×	0.194(0.057)	–0.001(0.022)	0.020(0.010)
$\hat{\Delta}_{DR}^{\beta_0}$	×	✓	0.015(0.038)	–0.038(0.018)	0.009(0.008)
$\hat{\Delta}_{DR}^{\beta_0}$	×	×	3.881(0.074)	3.704(0.036)	3.737(0.016)
$\hat{\Delta}_{CF}^{\beta_0, med}$	✓	✓	–0.066(0.091)	–0.032(0.019)	0.013(0.008)
$\hat{\Delta}_{CF}^{\beta_0, med}$	✓	×	–2.151(0.141)	–0.383(0.025)	–0.051(0.010)
$\hat{\Delta}_{CF}^{\beta_0, med}$	×	✓	0.014(0.047)	–0.038(0.018)	0.009(0.008)
$\hat{\Delta}_{CF}^{\beta_0, med}$	×	×	4.193(0.108)	3.726(0.036)	3.741(0.016)

Simulation 1: General performance

Estimator	Model		Sample size		
	PS	OR	$n = 50$	$n = 200$	$n = 1,000$
Root mean integrated squared error $\times 100$					
$\hat{\Delta}_{OR}^{\hat{\mu}_0}$	–	✓	0.695(0.016)	0.339(0.008)	0.150(0.003)
$\hat{\Delta}_{OR}^{\hat{\mu}_0}$	–	×	2.941(0.05)	2.773(0.027)	2.796(0.012)
$\hat{\Delta}_{IPW}^{\hat{\mu}_0}$	✓	–	2.912(0.108)	0.981(0.029)	0.425(0.009)
$\hat{\Delta}_{IPW}^{\hat{\mu}_0}$	×	–	3.176(0.057)	2.715(0.026)	2.736(0.012)
$\hat{\Delta}_{DR}^{\hat{\mu}_0}$	✓	✓	0.740(0.017)	0.348(0.008)	0.156(0.004)
$\hat{\Delta}_{DR}^{\hat{\mu}_0}$	✓	×	0.991(0.028)	0.409(0.010)	0.183(0.004)
$\hat{\Delta}_{DR}^{\hat{\mu}_0}$	×	✓	0.709(0.016)	0.342(0.008)	0.150(0.003)
$\hat{\Delta}_{DR}^{\hat{\mu}_0}$	×	×	2.942(0.048)	2.712(0.026)	2.736(0.012)
$\hat{\Delta}_{CF}^{\hat{\mu}_0, med}$	✓	✓	0.981(0.054)	0.352(0.008)	0.156(0.004)
$\hat{\Delta}_{CF}^{\hat{\mu}_0, med}$	✓	×	2.163(0.107)	0.495(0.013)	0.188(0.004)
$\hat{\Delta}_{CF}^{\hat{\mu}_0, med}$	×	✓	0.834(0.047)	0.343(0.008)	0.150(0.003)
$\hat{\Delta}_{CF}^{\hat{\mu}_0, med}$	×	×	3.289(0.070)	2.728(0.026)	2.738(0.012)

Coverage probabilities of SCBs at nominal level 95% ($n = 50, 200, 1,000$):

- DR: 88.4%, 91.5%, 92.4%
- CF: 80.8%, 91.4%, 92.4%

Simulation 1: General performance

Observations.

- The proposed DR and CF estimators exhibit double-robustness, whereas the standard OR and IPW estimators do not.
- When models are correct, $(\text{SE of OR}) < (\text{SE of DR}) < (\text{SE of IPW})$.
 - Similar behaviour observed in the classical setting with real-valued outcomes.
- Although the CF estimator is more appealing theoretically, the DR estimator has better finite sample performance, especially when n is small.
- The coverage probabilities are reasonably close to the nominal level, considering the difficulty to derive effective confidence bands for functional objects.

Simulation 2: Data-adaptive estimators

In the second simulation, we investigate the performance of the DR and CF estimators when data-adaptive methods are used to estimate the nuisance functions.

Data generation:

- Scenario 1: Same as in Simulation 1
- Scenario 2: Both the OR and PS models are nonlinear (the X terms is replaced with $\sin(\pi X)$)

Model fitting: Smoothing splines with data-adopted tuning parameters.

- OR: Smoothing splines
- PS: Logistic smoothing splines

Simulation 2: Data-adaptive estimators

Results. 1,000 Monte Carlo replicates for each setting.

Estimator	Scenario	Sample size		
		$n = 50$	$n = 200$	$n = 1,000$
Bias of difference in medians $\times 100$				
$\hat{\Delta}_{DRA}^{\hat{\beta}_0}$	1	-0.047(0.127)	-0.096(0.046)	0.012(0.008)
$\hat{\Delta}_{DRA}^{\hat{\beta}_0}$	2	-0.205(0.139)	-0.010(0.042)	0.030(0.008)
$\hat{\Delta}_{CFA}^{\hat{\beta}_0}$	1	-2.505(0.534)	-0.469(0.035)	-0.102(0.011)
$\hat{\Delta}_{CFA}^{\hat{\beta}_0}$	2	-1.206(0.788)	-0.808(0.122)	0.180(0.021)
Root mean integrated squared error $\times 100$				
$\hat{\Delta}_{DRA}^{\hat{\beta}_0}$	1	1.604(0.091)	0.545(0.044)	0.156(0.004)
$\hat{\Delta}_{DRA}^{\hat{\beta}_0}$	2	1.867(0.096)	0.686(0.153)	0.161(0.004)
$\hat{\Delta}_{CFA}^{\hat{\beta}_0}$	1	3.648(0.355)	0.767(0.024)	0.244(0.006)
$\hat{\Delta}_{CFA}^{\hat{\beta}_0}$	2	5.111(0.894)	2.202(0.183)	0.568(0.012)

Observations.

- The data-adaptive methods work reasonably well, although not as well as when both models are correctly specified parametrically.
- The RMISE of the data-adaptive methods decay as the sample size increases.

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Data application

The proposed methodology is applied to estimate the *causal effect of marriage on physical activity levels*.

In the 2005-2006 cycle of NHANES, participants were asked to wear an Actigraph for 7 days to monitor their physical activity levels.

- Physical activity intensity is measured in counts per minute (cpm), ranging from 0 to 32,767.
- For each individual, the physical activity intensity was recorded every minute for 24 hours, over 7 days.

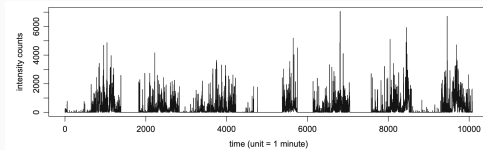


Figure reference: Chang & McKeague (2022).

Data application

- However, the trajectory of activity intensity is not directly comparable across different subjects as different individuals might have different circadian rhythms.
- Instead, the *distribution of activity intensity* is invariant to circadian rhythms, and hence can be used for making comparisons.

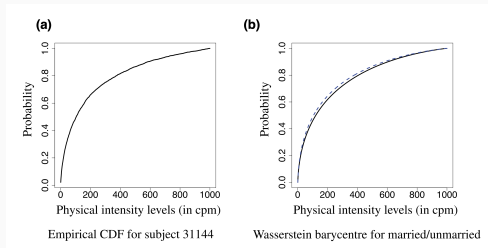


Figure 2. Panel (a) plots the preprocessed data for a randomly selected subject. Panel (b) plots the Wasserstein barycentre of the empirical cdfs in the married group (black solid line) and unmarried group (blue dashed line).

Variables:

- Treatment: Marital status ($A = 1$ if married, 0 otherwise)
- Outcome: Distribution of physical activity intensity over 7 days
- Covariates: Age and gender

Data summary:

- Sample size: 5,524 participants (after preprocessing)
- Marital status: 2,682 married, 2,842 unmarried
- Age: Mean 40.2 years (SD 21.3 years)
- Gender: 53.5% female

Result 1. Estimates of the mean potential outcomes

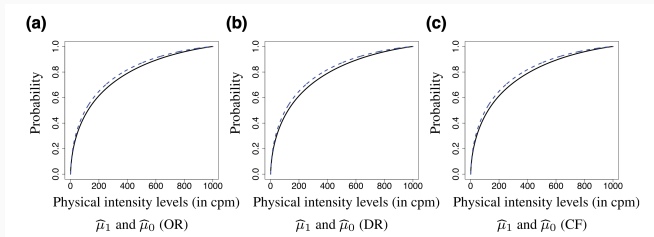


Figure 3. Panel (a), (b), and (c) plots $\hat{\mu}_1$ (black solid line) and $\hat{\mu}_0$ (blue dashed line) estimated by OR, DR, and CF, respectively.

- $\hat{\mu}_a$ is the corresponding cdf of the estimate $\hat{\mu}_a^{-1} = \hat{\mu}_a^{-1, \hat{\lambda}} \circ \lambda^{-1}$.
- The estimates of $\hat{\mu}_1$ are stochastically greater than those of $\hat{\mu}_0$, suggesting that marriage improves the entire distribution of physical intensity level at every quantile.

Result 2. Estimates of causal effect maps (difference in quantiles)

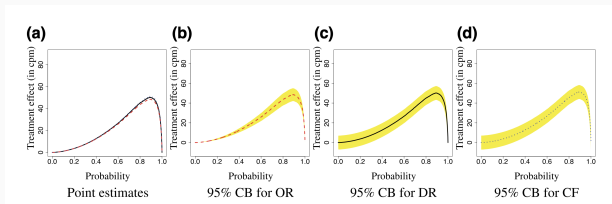


Figure 4. Difference in quantiles estimates with the OR (red dashed), DR (black solid), and CF (blue dotted) estimators: point estimates and 95% confidence bands.

- The SCBs are computed using the asymptotic properties of the estimators.
- The estimation results from three methods are close to one another, with the OR method having the tightest confidence band.
- For instance, according to the DR estimation results, marriage improves the median physical intensity level by 19.1 cpm (95% CI: [12.1, 26.1]).
 - This can be interpreted on both the population level and individual levels.

Result 3. Wasserstein distance between the mean potential outcomes

- $W_2(\hat{\mu}_1^{DR}, \hat{\mu}_0^{DR}) = \|\hat{\Delta}^{\hat{\mu}_0^{DR}}\|_{\hat{\mu}_0^{DR}} = 27.6 \text{ cpm.}$

Result 4. Individual causal transport map and its application

- Suppose we want to estimate the unobserved potential outcome $Y_i(1)$ for a particular individual.
- For instance, consider Subject 31144, who was unmarried so that $Y_{31144} = Y_{31144}(0)$.
- First, we estimate the *individual causal effect map* using the average causal effect map with reference distribution $\hat{Y}_{31144}$.
- Then, we apply the *individual causal transport map* to his empirical cdf to obtain $\hat{Y}_{31144}(1)$.
 - Recall that the individual causal transport map can be estimated by the relation $\hat{T}_{31144} = \hat{\Delta}^{\hat{Y}_{31144}} + \text{id.}$

Result 4. Individual causal transport map and its application

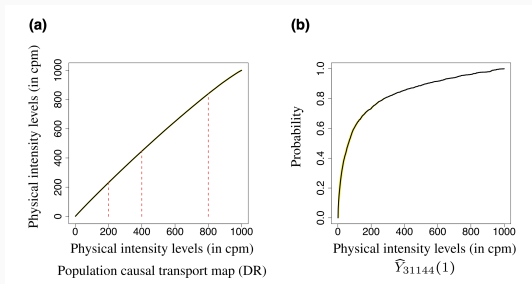


Figure 5. Panel (a) plots the DR estimate of population causal transport map, superimposed with the 95% confidence band; Panel (b) plots the DR estimate of counterfactual outcome $Y(1)$ with 95% confidence band for individual 31144.

- One may obtain that, for instance, getting married would raise his median physical activity intensity from 144.5 to 166.3 cpm (95% CI: [159.5, 173.7]).

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Contributions of the paper:

- (i) Proposed novel definitions of causal effects in a Wasserstein space and developed corresponding estimators, along with an analysis of their asymptotic properties.
- (ii) This is the first formal study of causal effects for outcomes defined in a *nonlinear space*.

Future directions:

- (i) Extending causal inference to Wasserstein spaces of probability distributions on higher-dimensional Euclidean spaces.
- (ii) Generalizing the approach to elements on Riemannian manifolds.
- (iii) Developing efficiency theory for distribution-valued outcomes.

-  Chang, H.-w., & McKeague, I. W. (2022). *Empirical likelihood-based inference for functional means with application to wearable device data*. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 84(5), 1947–1968.
-  Lin, Z., Kong, D., & Wang, L. (2023). *Causal inference on distribution functions*. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(2), 378–398.
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